

TEACHER GUIDE

Using this workbook

Farm in a Box is a Stage 4 Technology paper companion to the guided course. Readings teach the thirty named sections; selected source questions and adapted paper investigations develop formative evidence. The project folio collects decisions, tests, observations and evaluation.

Print and use

The student book has 50 A4 pages including cover and contents: 25 sheets printed double-sided, flipping on the long edge. Use the PDF for controlled printing. The Word edition keeps text, questions, tables and working fields editable; check pagination again after local edits.

Sequence and evidence

The course brief sets 25 × 60-minute periods across approximately ten weeks. Follow the published module order and current local teaching program. Teacher-led slides, videos, demonstrations and approved practical work can accompany paper records. The book does not replace teacher observation or create a formal marking scheme.

Confirm before practical work

Confirm crop, growing location, care access, materials, tools, processes, budget, PPE and current local controls. Use the school's current SOP/SDS and demonstrations. Animals, aquaponics, mains-powered systems, pesticides and fertiliser recipes are not assumed. Measuring or photographing a harvest does not approve eating it; consumption requires separate teacher-led food-safety and allergy controls.

Judge the reasoning

Answers below are keyed to student pages. Source MCQs have one best answer and a rationale. Open tasks accept alternative defensible responses. Check whether a claim names relevant evidence, recognises uncertainty and leads to an appropriate next action. Never turn a model response into a fabricated student result or teacher sign-off.

When growth or access is disrupted

A failed crop, delayed growth or missing observation can still support sound diagnosis and redesign. Provide labelled photographs, an approved specimen or a clearly attributed fallback record where needed. Students must identify supplied or hypothetical evidence and must not invent measurements.

Course and authority

<https://stevenowell.github.io/Stage-4-Farm-in-a-Box/>

Course source revision: eca8f30711335904a5fe3ebf01bab0f764d316e0. Design brief v1.0 and current teacher directions control local implementation. Illustrations show concepts, not approved construction plans. No physical test print is claimed.

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Design and initial folio

Student page 3 · The design challenge

Write one question your team must resolve before selecting its design. Explain why the answer matters.

Accept a specific unresolved constraint such as weekend care, crop suitability or location, linked to a design consequence.

Student page 44 · Site and success criteria

Write your project problem statement: need, intended users and key limit. Keep it open to different solutions.

Names the need and intended users/context plus a genuine limit, without selecting a final container or water system prematurely.

Record the site, date, light or shade, access to water, security and one change you would expect during the day.

A dated site record distinguishes observation from prediction and identifies a relevant constraint.

Write at least four balanced criteria. For each, name what you will judge and the evidence you will collect.

At least four criteria cover relevant functions, resource awareness, manageability, responsible production or communication. Each has observable evidence; no invented marks or pass thresholds.

Teacher confirmation to record: approved crop, materials, site and care arrangements.

Records actual teacher-confirmed details, or identifies what is still unconfirmed. No invented approval.

Student page 45 · Two concepts and a decision

Concept A — annotate how it works.

A readable concept with meaningful functional annotations and approval points.

Concept B — show a genuinely different arrangement.

A functional alternative with clear water/support/access relationships, not a cosmetic duplicate.

Which concept best fits your criteria? Explain one trade-off and one refinement.

A reasoned choice supported by common criteria, with a limitation and visible next refinement.

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Module 01 Agriculture Is Everywhere

Course: <https://stevencowell.github.io/Stage-4-Farm-in-a-Box/modules/module-01.html>

Student page 4 · Agriculture is everywhere

Trace a cotton shirt backwards: identify its living source, one process and one example of feedback.

Cotton plant; fibre separated from seed or textile processing; a recorded quality or resource observation used to change a production decision. A list of resources alone is not feedback.

Student page 5 · From paddock to product

Choose packaging or distribution. Explain one value it adds and one possible impact worth measuring.

Packaging protects/contains/informs and uses materials or can create waste; distribution provides access and uses transport resources. Accept a supported pairing and a relevant measure, not a claim that the stage is always harmful.

Student page 6 · Mission: grow more with less

Write one checkable criterion and one constraint for a compact growing system. Label which is which.

Criterion describes observable success, for example water loss can be seen and recorded.

Constraint is a real limit such as allocated space or confirmed care time; exact local values require teacher confirmation.

Student page 7 · Apply your learning

Question 1

B. Riverina–Murray rice is a grain industry shaped by irrigation and water availability; NSW cotton is a fibre industry affected by seasonal and water conditions. — Correct. This identifies different product properties and source-backed production contexts.

Question 2

B. Does it protect the product, and what material or waste does it add? — Correct. This balances the value provided with resource and waste considerations.

Question 3

B. Compare useful performance with the inputs, space, time and waste involved — Correct. Efficiency is a relationship that must be judged using evidence.

Choose rice or cotton. Write a short chain from plant to useful product. Identify an input, an intended output and a possible waste output. Explain one feedback record that could improve a stage, then write one checkable resource-use criterion for a small crop system.

Rice: grain harvested and milled for food; cotton: fibre separated from seed and prepared for textiles. Accept context-consistent inputs and outputs. Feedback must inform a change, and criterion must be observable. Do not invent water allocations or production quantities.

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Module 02 Country, Care and Future Farms

Course: <https://stevencowell.github.io/Stage-4-Farm-in-a-Box/modules/module-02.html>

Student page 8 · First foods: plants, animals and nutrition

Name the plant case and animal case. State one claim these public accounts do not authorise.

Plant: Wadawurrung sovereign grasses, with the named Traditional Owners corporation and CSIRO.

Animal: Gunditjmara kooyang at Budj Bim. They do not authorise copying cultural practice, invented nutrient data or universal claims about all First Nations communities.

Student page 9 · Caring for Country and future farms

Give one careful similarity and two important differences between the Wadawurrung partnership and WaterWise. Attribute each example to its named context.

Both use observation/evidence to inform resource decisions. Wadawurrung case centres named Traditional Owner authority and grassland research; WaterWise concerns sensing/prediction for irrigation. Preserve context and avoid ranking systems or treating them as interchangeable.

Student page 10 · A sustainability lens

For a replaceable component, identify one environmental benefit and one economic question.

Benefit: less whole-system disposal or longer useful life. Economic question: replacement cost, availability, labour, durability or maintenance time. Accept other justified answers; no guaranteed benefit without evidence.

Student page 11 · Apply your learning

Question 1

A. The Wadawurrung sovereign-grasses case, the Gunditjmara Budj Bim kooyang case and a clearly bounded Torres Strait Islander public overview — Correct. Together the named sources provide plant, animal, nutrition and food-availability evidence while preserving context.

Question 2

A. Both use evidence, while their knowledge systems, governance and technologies remain distinct — Correct. It states a similarity and preserves significant differences.

Question 3

D. It compares resource effects, people and responsibility, affordability and performance over time — Correct. This combines the three lenses and a whole-life view.

Compare the Wadawurrung sovereign-grasses plant case and Gunditjmara Budj Bim kooyang animal case. For each, name the knowledge holders/place and what the public evidence supports about nutrition or food availability. Add how the Queensland Health Torres Strait Islander overview broadens the discussion. Finish with one claim you must leave unmade.

Wadawurrung Traditional Owners corporation/CSIRO: sovereign-grasses nutritional-potential research and grassland care. Gunditjmara/Budj Bim: kooyang aquaculture supporting availability, not nutrient numbers. Queensland Health broad overview includes garden and marine animal foods, season, health and energy; not a particular island's local Cultural Knowledge. No copying permission, universalisation or invented details.

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Module 03 Start a Living Crop

Course: <https://stevencowell.github.io/Stage-4-Farm-in-a-Box/modules/module-03.html>

Student page 12 · Plant parts that do the work

Sketch a plant from the illustration. Label roots, stem and leaves, and link each to its function. Add a visible reproductive structure only if present.

Labels must identify the correct structures. Roots anchor and absorb water/minerals; stem supports and transports; leaves carry out photosynthesis. Flowers/fruits are labelled only when actually present. Accept a simplified, scientifically coherent diagram.

Explain why a damaged root system can affect leaves, even when the leaves receive light.

Roots supply water and dissolved minerals; connected transport supports leaves and photosynthesis. Light alone is insufficient. Damage is a possible explanation needing evidence, not a diagnosis from appearance alone.

Student page 13 · Germination and crop choice

Why might the fastest-growing crop still be a poor choice for this project?

It may not suit confirmed light/temperature/water, space, intended product or care opportunities. Match multiple criteria to context; speed alone does not establish suitability.

Student page 14 · Sow safely

A classmate wants to substitute an unfamiliar growing material. What should happen before work continues?

Pause and seek teacher confirmation; use only approved materials and the current local controls. Do not assume an online picture, peer suggestion or written checkbox approves the substitution.

Student page 15 · Apply your learning

Question 1

D. Water is matter used and moved by the plant; sunlight supplies energy — Correct. This distinction prevents the common idea that plants eat light.

Question 2

A. Its verified needs and life cycle fit the confirmed conditions, time, space and project purpose — Correct. This matches crop evidence to the actual brief.

Question 3

A. Pause, make the area safe as directed and seek the teacher's instruction — Correct. Changed conditions are a clear stop point for local review.

Using a teacher-confirmed crop or supplied example, write its identity and the conditions it needs. Compare it with another approved or supplied crop option. Justify its fit with the project's time, space and care. Plan one dated baseline observation and a consistent follow-up observation. Finish by naming what the teacher must confirm before sowing. If no crop is confirmed, label the entry "proposal".

Identity and conditions must derive from teacher/reliable crop information; match context rather than speed alone. Baseline and follow-up should be dated and comparable. Approval includes actual materials, workspace, tools, demonstrated process and local controls. Do not record a practical result or teacher sign-off that did not occur. Compare the selected option with another confirmed or supplied option against the same time, space, conditions and care needs.

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Module 04 Soil, Media and Circular Nutrients

Course: <https://stevenccowell.github.io/Stage-4-Farm-in-a-Box/modules/module-04.html>

Student page 16 · What makes a growing medium work?

Why is “holds the most water” an incomplete reason for choosing a growing medium?

Roots also need oxygen, drainage and support; saturated pore spaces can restrict air. Selection must fit crop/container/care conditions and multiple properties, not maximum storage alone.

Student page 17 · Nutrients, compost and circular farming

Draw an arrow pathway from suitable plant residue through decomposers to an approved return to the growing system. Label a control and a loss.

Residue → managed decomposition by organisms → suitable compost/organic matter returned under local approval. Include a real control such as suitable inputs or handling, and a loss or external input. It is not a perfect closed loop and does not replace photosynthesis.

Explain one benefit and one limit of returning suitable plant residue through managed composting.

Potential benefit: organic matter/nutrients return to useful growing pathways and some waste is reduced. Limits include approval, suitability, decomposition time, labour, energy, losses and quality controls. Compost does not supply the plant's photosynthetic sugars.

Student page 18 · Run a fair media test

In a media drainage comparison, identify the independent variable and one useful controlled variable.

Independent: approved medium option. Control: same container size/design, equivalent amount of medium, same water input or same timing/measurement method, as appropriate to the teacher-approved method.

Student page 19 · Apply your learning

Question 1

D. It does not directly test every property, such as nutrient suitability or long-term support — Correct. Test conclusions must stay within what was measured.

Question 2

D. Processes still involve time, energy, labour, losses and quality controls — Correct. Circular design can reduce impacts without making them disappear.

Question 3

D. Connect the measured result to a relevant criterion and weigh it with other evidence — Correct. A single property contributes to, but does not automatically settle, the decision.

Plan a comparison of two teacher-approved media for one property. State the question, independent and dependent variables, two controlled conditions, measuring units and a repeat. Explain which design criterion the result could inform and one conclusion it could not support. This is a paper plan until the teacher approves the method.

Accept a coherent drainage/water-holding or other sourced property test. Independent variable is medium; measured response matches question. Controls and repeat must be sensible, units relevant. Evidence informs one property, not universal quality, nutrient suitability or guaranteed yield. No unapproved practical recipe.

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Module 05 Water-Smart Growing

Course: <https://stevencowell.github.io/Stage-4-Farm-in-a-Box/modules/module-05.html>

Student page 20 · Where every drop goes

A reservoir level falls. Name two possible pathways and explain why another observation is needed.

Possible pathways include root uptake, evaporation, leakage/drainage or transfer into medium storage. Level change alone cannot separate them; inspect a relevant pathway or plant/root-zone condition. Avoid assuming all loss is transpiration or waste.

Student page 21 · Choose a water-smart idea

Compare manual watering and a passive wick using the same criterion: maintenance or visibility.

Manual watering needs repeated human attention but is readily observed; a wick may reduce active delivery but needs checks for contact/blockage/transfer. Compare both against the chosen common criterion and acknowledge context.

Student page 22 · Build and test a wick

A wick transfers no visible water. Propose a sensible next step and state what the test cannot prove.

Check contact/continuity and method, then change one approved variable while matching others and repeat measurement. A bench test cannot establish future crop yield or whole-system performance; record failure honestly.

Student page 23 · Apply your learning

Question 1

D. Account for water entering, being stored or used, and leaving by different pathways — Correct. This makes feedback and improvement opportunities visible.

Question 2

A. Automated systems still need suitable settings, maintenance, evidence and human oversight — Correct. Automation changes the work; it does not remove responsibility.

Question 3

B. This arrangement transferred water during the bench test; full-system performance still needs checking — Correct. It reports transfer and acknowledges the next evidence need.

For a hypothetical compact crop system, compare manual, drip and passive-wick delivery using maintenance and checking access. Select one idea for a prototype and justify the trade-off. Describe one observable failure point and one measurement that would help test it. Keep materials and local care arrangements labelled “Teacher to confirm”.

Manual: simple/visible, repeated human input. Drip: targeted delivery, layout/operation checks. Wick: passive, contact and transfer checks. Selection may vary if justified. Failure and measure should match selected concept; bench transfer does not prove harvest. Hypothetical evidence must not be presented as observed results.

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Module 06 Farm Under Pressure

Course: <https://stevenccowell.github.io/Stage-4-Farm-in-a-Box/modules/module-06.html>

Student page 24 · Read the microclimate

Why would two readings from one afternoon be weaker than records collected consistently across several days?

They may describe only one event; repeated comparable records better reveal a pattern. Accept references to consistent time, site and method.

Student page 25 · Diagnose crop stress

Write one symptom a student can observe and one question that should be investigated before naming its cause.

Any source-supported symptom, such as drooping leaves, paired with a question about timing, distribution, recent changes or comparison conditions. Symptom must not be presented as proven cause.

Student page 26 · Biosecurity and responsible choices

An unfamiliar mark appears on several leaves. Give your first response and explain one consequence it helps prevent.

Pause; observe and record without unnecessary handling; report and follow directions. Prevents spread or loss of useful evidence; no invented identification or treatment.

Student page 27 · Investigate crop stress

Question 1

D. The same variable recorded with units at stated times — Correct. A consistent method and timing make a later pattern easier to detect; see the linked theory.

Question 2

D. A production record showing when the symptom and recent changes occurred — Correct. Timing and history can support or weaken an explanation; see the linked theory.

Question 3

C. To reduce the movement and impact of harmful pests, diseases or contaminated material — Correct. Biosecurity is shared care that limits spread and impact; see the linked theory.

Fictional case: on two visits, leaves in one box droop while a nearby box appears unchanged. The record does not show recent care or site conditions. An unfamiliar mark is also visible. Use arrows: observation → two possible explanations → evidence needed to distinguish them. Identify the missing records. Finish with the responsible next action; do not prescribe a treatment.

Accept multiple plausible hypotheses stated as possibilities, not diagnoses. Require observation/cause separation, relevant comparison and production records, and stop–record–avoid spread–report response. Teacher directs handling; no invented treatment. The pathway links each hypothesis to a discriminating observation or record rather than jumping to treatment.

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Module 07 Design the Farm in a Box

Course: <https://stevencowell.github.io/Stage-4-Farm-in-a-Box/modules/module-07.html>

Student page 28 · Turn the brief into criteria

Improve the criterion “easy to monitor” so another student could check it.

Example: observations can be recorded without moving the growing area, checked using a teacher-approved observation task. Accept other observable features; no invented required dimensions.

Student page 29 · Generate, compare and refine

Why should a concept with the highest matrix total still be examined for weaknesses?

Totals can hide weak criteria or unsupported assumptions. Examine evidence, trade-offs and important risks before refining the concept.

Student page 30 · Plan the build and control risk

Name one dependency and explain why a checkpoint belongs before the next task.

Example: system function/stability checks before establishment; catching a weakness prevents later work depending on an unsuitable system. Approved local plan controls exact sequence.

Student page 31 · Compare three design concepts

Question 1

A. The system must work within the approved small space — Correct. It sets a boundary within which the solution must work; see the linked theory.

Question 2

A. The relative importance of a criterion — Correct. It shows why some criteria influence the decision more; see the linked theory.

Question 3

B. To check basic function and stability before later work depends on them — Correct. The pause prevents an early weakness travelling forward; see the linked theory.

Draw a third, functionally different concept here. Compare it with concepts A and B on page 45 using the same two criteria and weights. Explain the trade-off, chosen refinement and approval still needed.

Three concepts differ in function or arrangement. Apply the same criteria, scoring scale and weights to all. Explain weights before scoring; a high total requires evidence rather than taste. Justify a refinement and retain teacher approval for practical details.

Concept C with functional annotations

A third arrangement shows root zone, water pathway, access and portability; annotations explain functional differences from A and B.

Concept

Choose the same criteria and a consistent scale, record weights, and multiply each score by its criterion weight. Total each row. Explain why the weighting fits the brief; do not invent performance evidence.

Why these weights? Which trade-off and refinement does the evidence support?

Weights reflect intended users and the priority criteria. The selected concept and refinement follow evidence and recognise a disadvantage or uncertainty.

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Module 08 Build and Establish

Course: <https://stevencowell.github.io/Stage-4-Farm-in-a-Box/modules/module-08.html>

Student page 32 · Ready the workspace

A bench is tidy but a resource differs from the demonstrated one. Which gate decision is justified, and why?

Stop and ask; record the mismatch and obtain teacher direction. Tidiness cannot approve a substitution or its controls.

Student page 33 · Build and test the box

What must a before-and-after test keep consistent, and why?

Keep the test method and relevant conditions consistent; modify one approved feature where possible. This makes the evidence comparable and limits competing explanations.

Student page 34 · Establish the crop

Why does an establishment photograph need a date, position and written note?

These connect the image to a particular starting condition and method, enabling later comparison. A photograph alone does not prove successful growth.

Student page 35 · Control a production checkpoint

Question 1

B. Ready, correct and recheck, or stop and ask — Correct. Each result leads to an explicit next action; see the linked theory.

Question 2

B. The teacher-approved plan for the real design — Correct. Tests must fit the local system and controls; see the linked theory.

Question 3

D. Date or stage, supplied material, location, initial condition and completed process — Correct. These details support later comparison; see the linked theory.

Fictional checkpoint: the approved functional test does not meet its stated pass condition. Record the function tested, the evidence you would preserve and the decision to stop or seek approval for one controlled change. Describe how a re-test could be comparable and what must be confirmed before crop establishment. Do not invent a tool, pass value or making procedure.

Require work–check–record decision cycle; contextual evidence; approved single modification or justified stop; same re-test method; functional readiness and approved establishment method before progressing.

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Module 09 Grow, Measure and Improve

Course: <https://stevencowell.github.io/Stage-4-Farm-in-a-Box/modules/module-09.html>

Student page 36 · Build a useful data record

A graph rises after a change. What can you say confidently, and what remains uncertain?

The recorded variable rose after the change; causation remains uncertain without supporting records and consideration of other conditions.

Student page 37 · Maintain, test and improve

What is missing from “We changed the system and finished”?

The trigger/evidence, approved single change, comparable later check and result. A recorded action alone cannot demonstrate improvement.

Student page 38 · Grow for community, waste less

Use a teacher-supplied current community growing initiative. Record its name, source and date; map actors, beneficiaries, inputs and waste. Propose a source-backed school connection and one measurable waste-prevention decision. A hypothetical map is preparation only.

Use a named current initiative and record source/date. Identify contributors, beneficiaries, relevant inputs and a supported waste pathway. Propose a plausible school connection and link waste prevention to a measure. Distinguish a proposed connection from an actual partnership. Hypothetical mapping is preparation, not completed current-case evidence; no distribution or donation is approved by the worksheet.

Why is “we will donate every harvest” an unsuitable automatic plan?

Distribution depends on approved food-safety, allergy, quality and site pathways; availability and suitability are not guaranteed. Seek confirmation rather than promise donation.

Student page 39 · Graph and challenge a claim

Question 1

B. Mark it honestly as missing or not observed — Correct. This preserves integrity and lets the limit be considered; see the linked theory.

Question 2

B. It is linked to a recorded pattern, criterion or system check — Correct. The action responds to evidence rather than impulse; see the linked theory.

Question 3

A. Whether participation and access are fair — Correct. This concerns effects on people and inclusion; see the linked theory.

Use a real class record or a labelled teacher-supplied record. Plot one variable against time or condition; label axes, units and scale. Below, state a pattern, a limit, and one approved investigation with a comparable re-check and waste-prevention purpose. If evidence is unavailable, set up labelled axes and request the record; do not invent results.

The graph matches a traceable record, with suitable consistent scales, labelled variables/units and accurately plotted values. Explain the pattern without unsupported causation. State a limitation, teacher-approved single-factor investigation and comparable re-check connected to avoiding waste. Empty labelled axes are preparation, not completed results.

Graph title and labelled axes

A meaningful title and labelled axes include units and a suitable scale. Plot actual/supplied values consistently; label the record source and any missing values.

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Module 10 Harvest, Judge and Share

Course: <https://stevencowell.github.io/Stage-4-Farm-in-a-Box/modules/module-10.html>

Student page 40 · Harvest and trace the product

What different evidence can a batch record and a temperature sensor provide?

Batch records trace identity and movement; a sensor can reveal temperature events (and location in the described example). Neither guarantees quality.

Student page 41 · Judge quality with evidence

When is “not yet demonstrated” fairer than “failed”?

When no valid opportunity or sufficient evidence exists to test a criterion. Explain evidence gap and what would establish performance.

Student page 42 · Pitch the next version

Turn “the next version will definitely work” into an evidence-led proposal.

Name a proposed feature change linked to an evidenced limitation, describe expected benefit conditionally, and name a test. Avoid guaranteed outcomes.

Student page 43 · Pitch the next version

Question 1

C. Stage, product or sample, approved method, quantity or quality evidence and loss — Correct. These details make the record traceable; see the linked theory.

Question 2

D. A documented access test and user observation — Correct. The evidence directly addresses the criterion; see the linked theory.

Question 3

C. ‘The input log shows a reduction after the approved modification, supporting one resource criterion in this test context’ — Correct. It names evidence, criterion and boundary; see the linked theory.

Use your original criteria and strongest records to write a short pitch: need and design response; one claim supported by two evidence items; why that evidence matters; one limitation; and a proposed feature change with its expected benefit and test. Distinguish product quality from system quality. Present future benefits as possibilities, not completed results.

Require coherent claim–evidence–reasoning, named criteria and two relevant evidence items, distinction between output and system, honest limitation, and a testable next proposal with approval boundaries. Accept negative or disrupted crop results supported by honest records.

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Plan and test evidence

Student page 46 · Plan production and control risk

Step and dependency

A sensible sequence, actual approved resources and checks before irreversible or dependent steps. Also require the attached dated production traveller: ready, work completed, check, actual evidence and next-authorisation status at each stage. Distinguish plans, real completed work and supplied fallback records. Record teacher observation or authorisation only when it actually occurred.

Identify two relevant hazards. Record the teacher-approved controls and the point at which you must stop and ask.

Hazards fit the actual activity; controls match local instruction. Stop for unexpected damage, spill, unfamiliar material or a changed procedure.

Who does what, and how will every person keep evidence of the reasoning?

Roles allow all students to understand the design, observations and decisions; handover is clear.

Student page 47 · Test one relationship

Name the test and evidence source. What question, prediction, changed variable, measured response and controls are appropriate?

A testable question with one changed variable, relevant measured response and appropriate controlled variables.

Trial or date

Honest repeated records, clear units and conditions; missing observations labelled instead of invented.

What does the evidence support? State a limitation and next check. For media, justify a medium choice; for the wick, justify one change and comparable retest.

Require two completed records, here and on the named attached sheet. Media: compare one tested property with criterion needs and justify a choice without overclaiming nutrient quality or yield. Wick: identify transfer evidence, one limitation, one controlled modification and comparable retest. Attribute fallback data; never invent results.

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Care harvest and evaluate

Student page 48 · Establish and care for the crop

Baseline: date, crop identity, starting condition and measurement method.

Includes crop and system identity, date and comparable observation or measurement; no assumed healthy baseline.

Date

Dated entries link the trigger, approved action and follow-up; condition scales or units are consistent.

Care handover: what needs attention before the next visit, and who has confirmed the arrangement?

A specific manageable handover; weekend or absence care is teacher-confirmed.

Student page 49 · Trace and judge the harvest

Record item

Traceable records identify the crop/system, date, maturity observations, handling and authorised destination. Fallback data is labelled honestly.

Trace the wider product chain with verbs and arrows. Mark stages confirmed, hypothetical, not applicable or awaiting approval.

Connect relevant production, harvest, transformation, packaging, distribution and use stages. Not every crop needs every stage. Label hypothetical/unused/unapproved stages and retain traceable crop/system identity; do not claim eating or donation is authorised.

Judge two quality features using observations or measurements. What can you conclude, and what remains uncertain?

Judgements use observable quality features and acknowledge limitations; no unsupported food-safety or edibility claim.

How could one avoidable input or waste pathway be reduced next time?

A specific prevention or approved reuse/recovery proposal linked to the actual record.

Student page 50 · Evaluate and pitch the next version

Return to your criteria on page 44. What does your evidence show for each?

Each judgement connects a criterion to actual evidence; distinguishes success, partial success and untested criteria.

Propose one specific change. Why should it help, what approval is needed, and how would you test it?

A feasible next version follows evidence, identifies local approval and a comparable retest.

Prepare a short field-day pitch: need, design choice, strongest evidence and next version.

A coherent explanation gives the audience a relevant need, decision, evidence and specific next step; avoids exaggerated claims.